PROJECT

Socio-Economic Drivers of Politcal Polarisation in Germany

#Team

Data Smiths

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SUMMARY

The study examines how socio-economic and demographic variations across Germany's 16 federal states influence party vote shares. Using statistical techniques such as regression analysis,Scatter plots, Geo maps, and Correlation Matrix, it identifies which regions and voter groups have the greatest impact on electoral outcomes, highlighting key drivers of vote shifts.

INTRODUCTION

Elections are shaped not only by political campaigns but also by the underlying socio-economic and demographic characteristics of voters. In Germany, significant variation exists across the 16 federal states in terms of income, education, age, and urbanization, all of which can influence party preferences. Understanding how these factors affect vote shares is crucial for identifying which regions and voter groups are most pivotal in determining electoral outcomes. This study employs statistical techniques, including regression analysis, to examine the relationship between regional socio-demographic profiles and party support, shedding light on the key drivers behind shifts in the political landscape.

PROBLEM

analyse how socio-economic and demographic differences across Germany's 16 federal states drive party vote shares and which regions and voter groups are most influential in shifting electoral outcomes

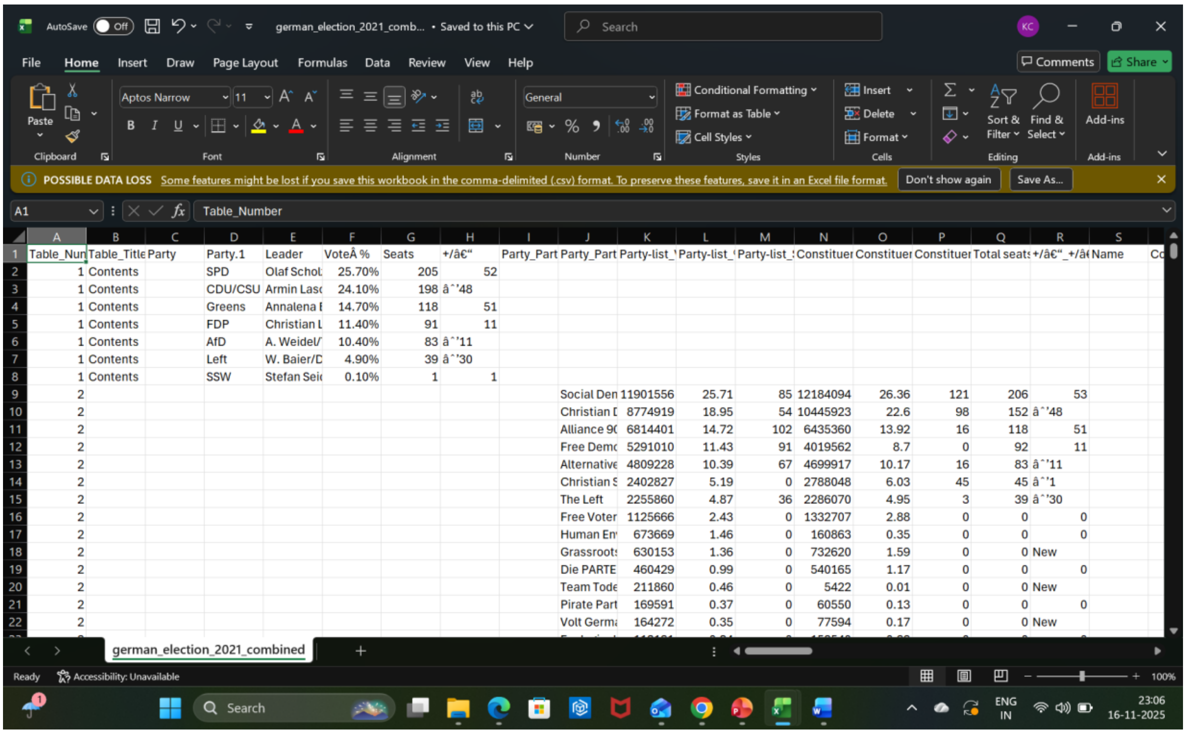
DATA

#Data cleansing

1.Gather datasets: Election results by state, socio-economic indicators (income, unemployment, education, age distribution), and demographic data. We mainly obtained this data through publicly available information sources such as Bundeswahlleiter, Destatis, and Wikipedia. While gathering data we faced a few issues though with clear goal in sight we concentrated our search on the official sources. We were looking for the driving factors that affect every individual in Germany such as economic factors of GDP per capita and median income, and by evaluating this data we can make a correlation between vote share for different parties and their strongholds in different regions.

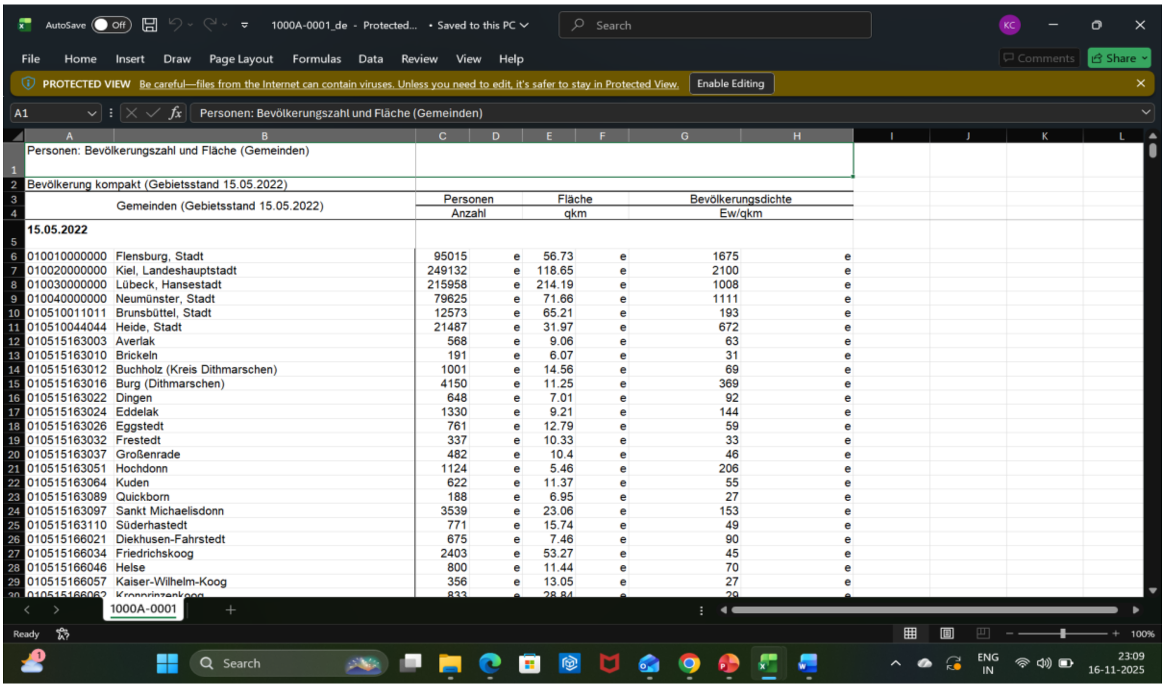
First and foremost we as a team decided to get the data for vote share of a particular party with regards to all the 16 states of Germany. Then, we filtered out most affecting factors in which we included GDP per capita, Median income, Tertiary education levels, Urbanization, Percentage of immigrants, and unemployment rate. Moreover, we added two more measures which are important because we wanted to categorize our find with different age-groups and voter turn-out.

Here I have attached a few screenshots and link of the websites from which we have gathered data.



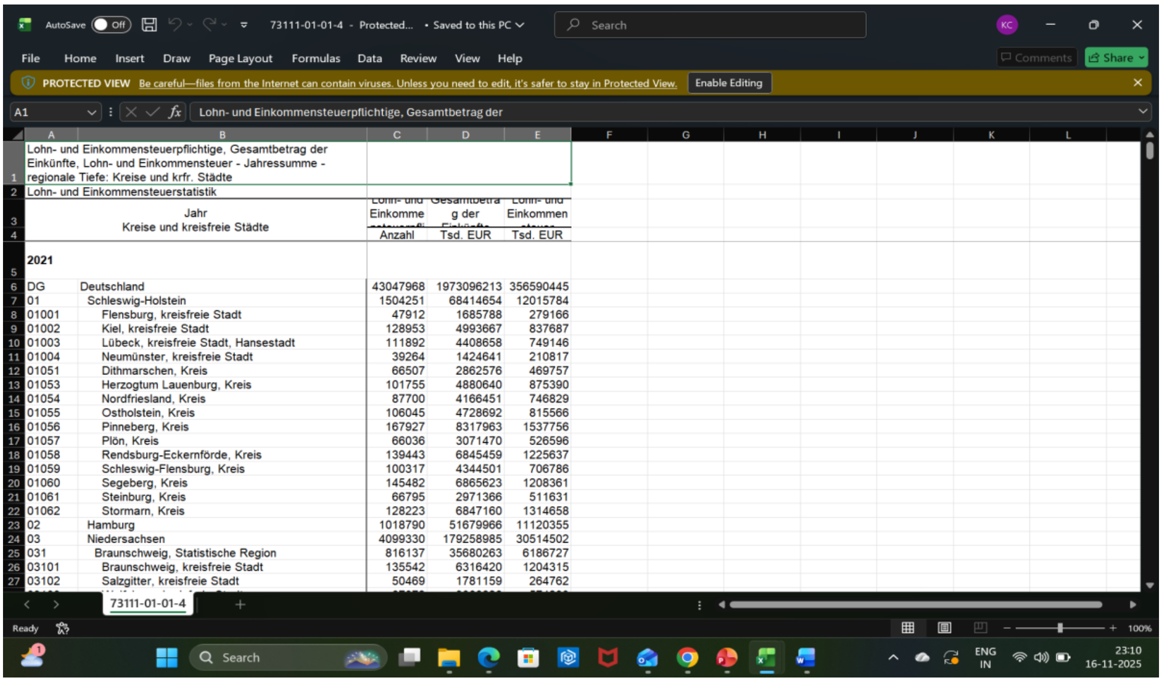
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
1	Table_Nun	Table_Title	Party	Party.1	Leader	VoteÂ %	Seats	+/â€"	Party_Part	Party_Part	Party-list_'	Party-list_'	Party-list_:	Constituer	Constituer	Constituer	Total seats	+/â€"_+/â€	Name	Cc
2	1	Contents		SPD	Olaf Schol	25.70%	205	52												
3	1	Contents		CDU/CSU	Armin Lasc	24.10%	198	âˆ'48												
4	1	Contents		Greens	Annalena I	14.70%	118	51												
5	1	Contents		FDP	Christian L	11.40%	91	11												
6	1	Contents		AfD	A. Weidel/	10.40%	83	âˆ'11												
7	1	Contents		Left	W. Baier/D	4.90%	39	âˆ'30												
8	1	Contents		SSW	Stefan Sei	0.10%	1	1												
9	2									Social Den	11901556	25.71	85	12184094	26.36	121	206	53		
10	2									Christian D	8774919	18.95	54	10445923	22.6	98	152	âˆ'48		
11	2									Alliance 90	6814401	14.72	102	6435360	13.92	16	118	51		
12	2									Free Demc	5291010	11.43	91	4019562	8.7	0	92	11		
13	2									Alternative	4809228	10.39	67	4699917	10.17	16	83	âˆ'11		
14	2									Christian S	2402827	5.19	0	2788048	6.03	45	45	âˆ'1		
15	2									The Left	2255860	4.87	36	2286070	4.95	3	39	âˆ'30		
16	2									Free Voter	1125666	2.43	0	1332707	2.88	0	0	0		
17	2									Human En	673669	1.46	0	160863	0.35	0	0	0		
18	2									Grassroots	630153	1.36	0	732620	1.59	0	0	New		
19	2									Die PARTE	460429	0.99	0	540165	1.17	0	0	0		
20	2									Team Tode	211860	0.46	0	5422	0.01	0	0	New		
21	2									Pirate Part	169591	0.37	0	60550	0.13	0	0	0		
22	2									Volt Germa	164272	0.35	0	77594	0.17	0	0	New		

Figure 1 Data derived from official government source



Personen: Bevölkerungszahl und Fläche (Gemeinden)							
Bevölkerung kompakt (Gebietsstand 15.05.2022)							
Gemeinden (Gebietsstand 15.05.2022)		Personen		Fläche		Bevölkerungsdichte	
		Anzahl		qkm		Ew/qkm	
15.05.2022							
010010000000	Flensburg, Stadt	95015	e	56.73	e	1675	e
010020000000	Kiel, Landeshauptstadt	249132	e	118.65	e	2100	e
010030000000	Lübeck, Hansestadt	215958	e	214.19	e	1008	e
010040000000	Neumünster, Stadt	79625	e	71.66	e	1111	e
010510011011	Brunsbüttel, Stadt	12573	e	65.21	e	193	e
010510044044	Heide, Stadt	21487	e	31.97	e	672	e
010515163003	Averlak	568	e	9.06	e	63	e
010515163010	Brickeln	191	e	6.07	e	31	e
010515163012	Buchholz (Kreis Dithmarschen)	1001	e	14.56	e	69	e
010515163016	Burg (Dithmarschen)	4150	e	11.25	e	369	e
010515163022	Dingen	648	e	7.01	e	92	e
010515163024	Eddelak	1330	e	9.21	e	144	e
010515163026	Eggstedt	761	e	12.79	e	59	e
010515163032	Frestedt	337	e	10.33	e	33	e
010515163037	Großenrade	482	e	10.4	e	46	e
010515163051	Hochdonn	1124	e	5.46	e	206	e
010515163064	Kuden	622	e	11.37	e	55	e
010515163089	Quickborn	188	e	6.95	e	27	e
010515163097	Sankt Michaelisdonn	3539	e	23.06	e	153	e
010515163110	Süderhastedt	771	e	15.74	e	49	e
010515166021	Diekhusen-Fahrstedt	675	e	7.46	e	90	e
010515166034	Friedrichskoog	2403	e	53.27	e	45	e
010515166046	Helse	800	e	11.44	e	70	e
010515166057	Kaiser-Wilhelm-Koog	356	e	13.05	e	27	e

Figure 2 Data derived from official source for socio-economic measures



Lohn- und Einkommensteuerpflichtige, Gesamtbetrag der Einkünfte, Lohn- und Einkommensteuer - Jahressumme - regionale Tiefe: Kreise und krfr. Städte				
Lohn- und Einkommensteuerstatistik				
Jahr Kreise und kreisfreie Städte		Lohn- und Einkomme	Gesamtbetra g der	Lohn- und Einkommen
		Anzahl	Tsd. EUR	Tsd. EUR
2021				
DG	Deutschland	43047968	1973096213	356590445
01	Schleswig-Holstein	1504251	68414654	12015784
01001	Flensburg, kreisfreie Stadt	47912	1685788	279166
01002	Kiel, kreisfreie Stadt	128953	4993667	837687
01003	Lübeck, kreisfreie Stadt, Hansestadt	111892	4408658	749146
01004	Neumünster, kreisfreie Stadt	39264	1424641	210817
01051	Dithmarschen, Kreis	66507	2862576	469757
01053	Herzogtum Lauenburg, Kreis	101755	4880640	875390
01054	Nordfriesland, Kreis	87700	4166451	746829
01055	Ostholstein, Kreis	106045	4728692	815566
01056	Pinneberg, Kreis	167927	8317963	1537756
01057	Plön, Kreis	66036	3071470	526596
01058	Rendsburg-Eckernförde, Kreis	139443	6845459	1225637
01059	Schleswig-Flensburg, Kreis	100317	4344501	706786
01060	Segeberg, Kreis	145482	6865623	1208361
01061	Steinburg, Kreis	66795	2971366	511631
01062	Stormarn, Kreis	128223	6847160	1314658
02	Hamburg	1018790	51679966	11120355
03	Niedersachsen	4099330	179258985	30514502
031	Braunschweig, Statistische Region	816137	35680263	6186727
03101	Braunschweig, kreisfreie Stadt	135542	6316420	1204315
03102	Salzgitter, kreisfreie Stadt	50469	1781159	264762

Figure 3 state data

Here are the links:

Bundeswahlleiter - Open Data (Bundestagswahl 2021 open data) for federal election data 2021

https://www.bundeswahlleiterin.de/bundestagswahlen/2021/ergebnisse/opendata.html

Destatis - Unemployment (regional), GDP(regional), Median income(Regional), Education level(regional)

https://www.destatis.de/EN/Themes/_node.html

Eurostat - regional statistics (optional) for age groups and percentage of foreign-born

https://ec.europa.eu/eurostat

2.Inspect data: Check for missing values, inconsistencies, or unusual entries (e.g., negative population values).

#Data transformation

Encoding categorical variables like states or parties into numerical formats (e.g., one-hot encoding) for statistical modeling.

Creating derived features such as voter turnout percentages or age-group shares to better capture patterns in voting behaviours.

Executed a python script to scrap and translate the raw data as most of the data was heavily loaded with the attributes that we didn't want.

Aggregating or scaling data at the state level to align socio-economic indicators with vote shares for accurate comparison.

Adding longitude and latitude to precisely showcase the data with 'Geo map' feature in SAS visual Analytics

#Data Enrichment

Data enrichment involved gathering additional socio-economic and demographic information through web scraping and processing all tabular entries with a Python script. The datasets were compiled into a master dataset, with new columns added to capture insightful variables for more robust analysis.

Here is the link for the final master table.

https://docs.google.com/spreadsheets/d/1IIN0Q3lD9RCjpI6oTUUm4YqWocQLMMKs/edit?usp=sharing&ouid=112454371391556562764&rtpof=true&sd=true

ANALYSIS

The analysis explored political voting behaviour in Germany using socio-economic and demographic indicators across all 16 federal states. By visually analysing patterns through geospatial charts, scatter plots, correlation matrices and regression modelling, several consistent and meaningful relationships emerged.

A central finding was the strong East–West divide, especially for the AfD. Eastern states recorded substantially higher AfD vote shares, and *Region* was confirmed as the single strongest predictor in the regression model. Economic hardship also played a significant role: states with higher unemployment rates consistently showed higher AfD percentages, while states with a larger foreign-born population showed a clear negative association with AfD support. This suggests that AfD's voter base is concentrated in economically strained, less diverse environments.

SPD displayed a different behavioural pattern. Although SPD performed across both sides of Germany, its support increased notably in states with medium to high unemployment. The geobubble and frequency charts highlighted that SPD gains were strongest in areas facing economic pressure, such as Bremen, Brandenburg and North Rhine-Westphalia. This pattern aligns with the party's traditional working-class foundation and appeal to voters prioritising welfare and employment security.

The Greens showed highly distinctive trends compared to other parties. Their strongest support came from states with high urbanisation, high educational attainment and higher income levels. Urban hubs like Berlin and Hamburg clearly illustrated this, showing both high tertiary education percentages and strong Green vote share. This demonstrates a demographic alignment between the Greens and younger, urban, university-educated voters.

CDU/CSU exhibited the opposite pattern: higher support in states with stronger economies, lower unemployment and older populations. Bavaria, Baden-Württemberg and Rhineland-Palatinate reflected this trend clearly, indicating that CDU/CSU's base lies in stable, high-income regions with conservative demographic structures.

Combining all variables across dashboards revealed a broader structure to Germany's electoral landscape.

- **Economically vulnerable and less diverse eastern states tended toward AfD and Left.**
- **Urban, highly educated western states leaned strongly toward Greens.**
- **CDU/CSU dominated in prosperous, ageing regions.**
- **SPD maintained a broad geographic distribution but strengthened in states facing unemployment challenges.**

Regression modelling further validated these relationships. Region, unemployment rate and foreign-born share were the most influential drivers affecting AfD vote share, while income, urbanisation and education played moderate but consistent roles. The model showed high predictive accuracy, proving that socio-economic factors are strongly connected to political outcomes in Germany.

Overall, the analysis demonstrated that voting patterns are not random but highly structured around economic security, demographic composition and regional identity. The study successfully identified the voter characteristics and regional conditions most associated with specific political parties, offering a clear and evidence-based understanding of why different regions in Germany vote the way they do.

Here are the screenshots of all the analysis done:

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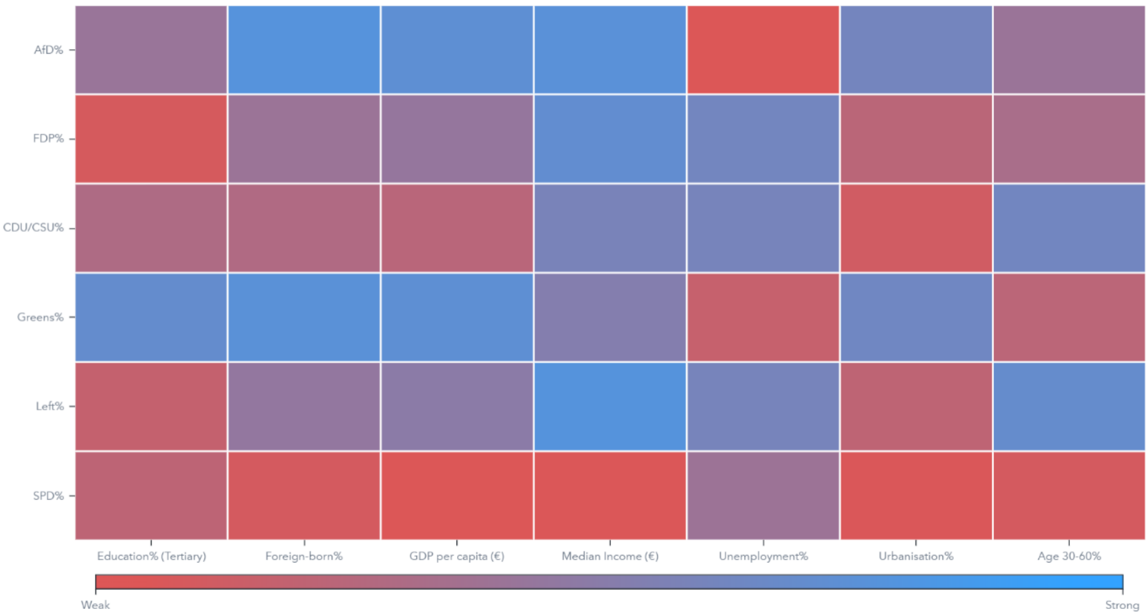
Correlation of Selected Measures



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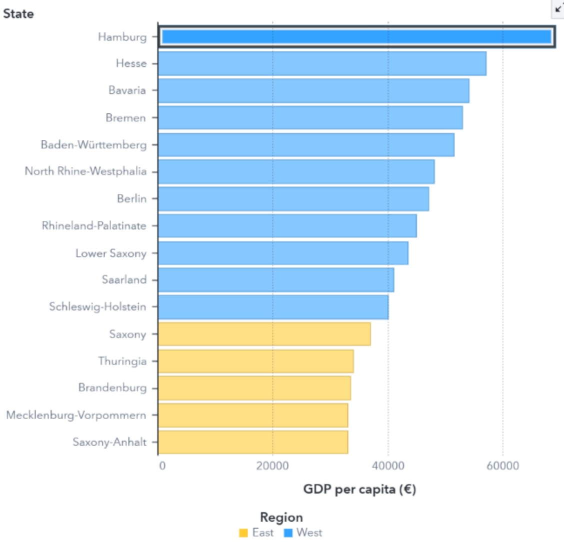
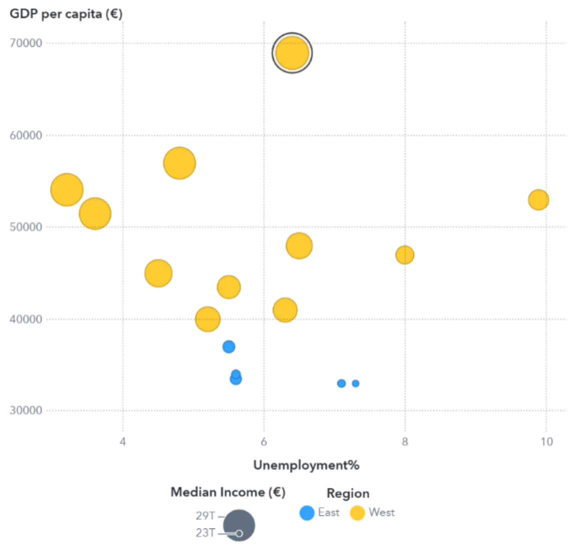
Economic Prosperity Shapes Germany's Regional Identity: Economic strength remains concentrated in the West. These inequalities set the stage for different political preferences we'll see next.

GDP per capita (€) by State grouped by Region



GDP per capita (€) by Unemployment% sized by Median Income (€)



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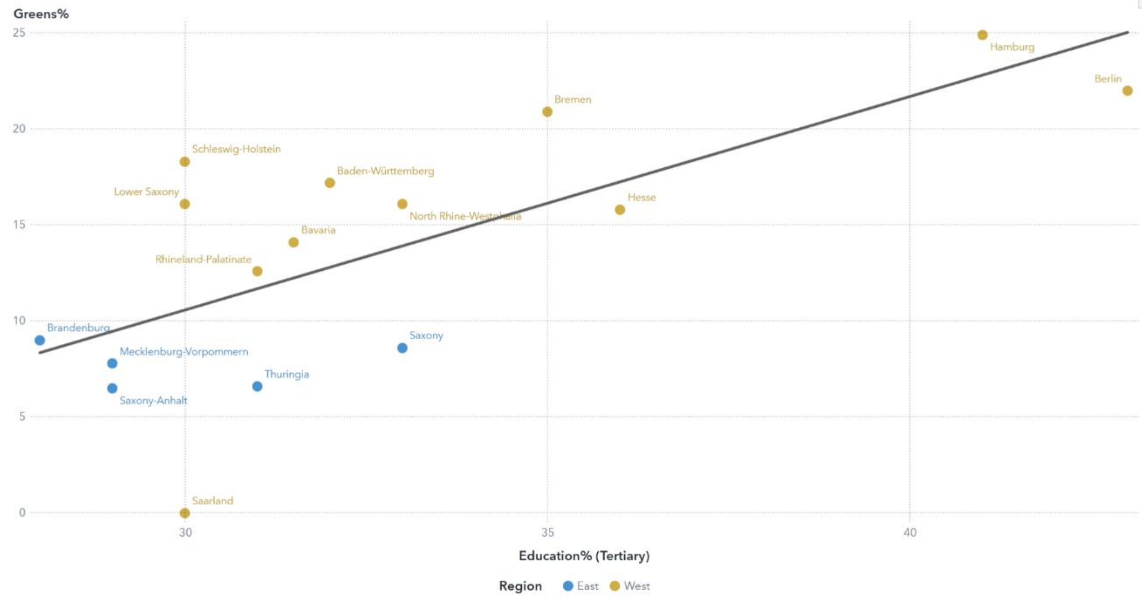
Scatter Plot of Selected Measures

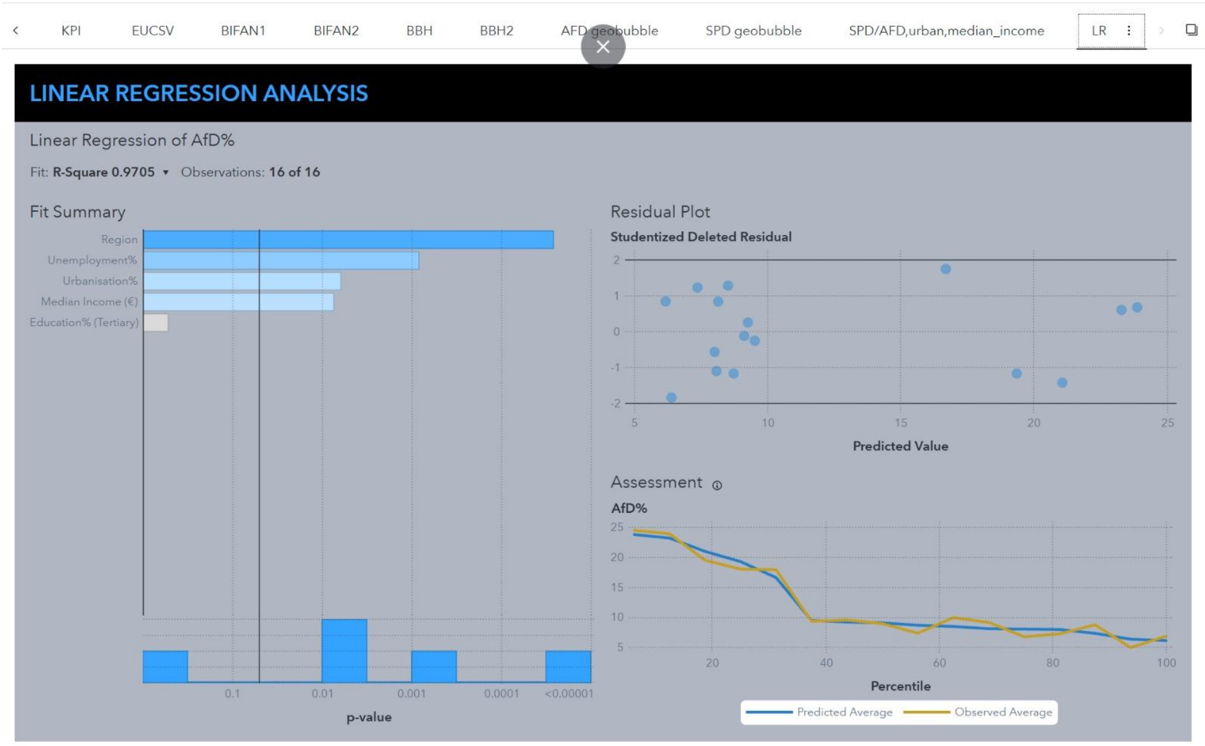
Scatter Plot of Selected Measures

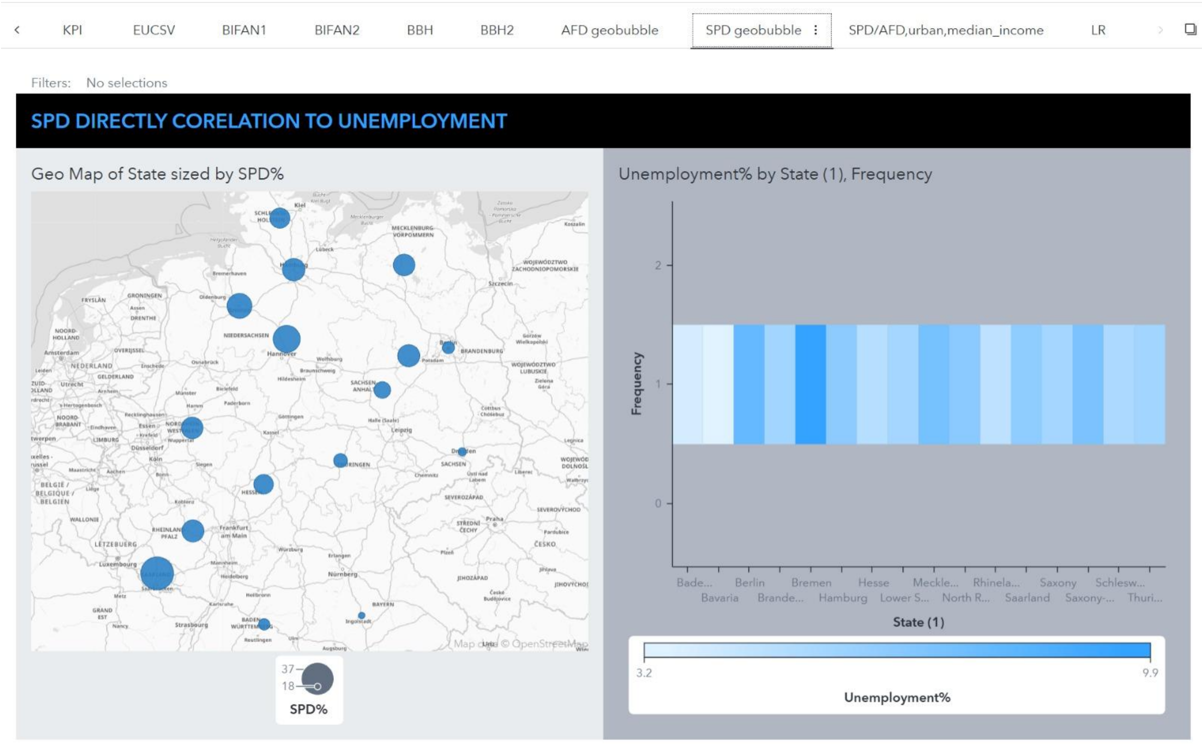
Page 3 Page 5 Page 6 Page 7

Green Party Vote Share vs Education (Tertiary, 2021)

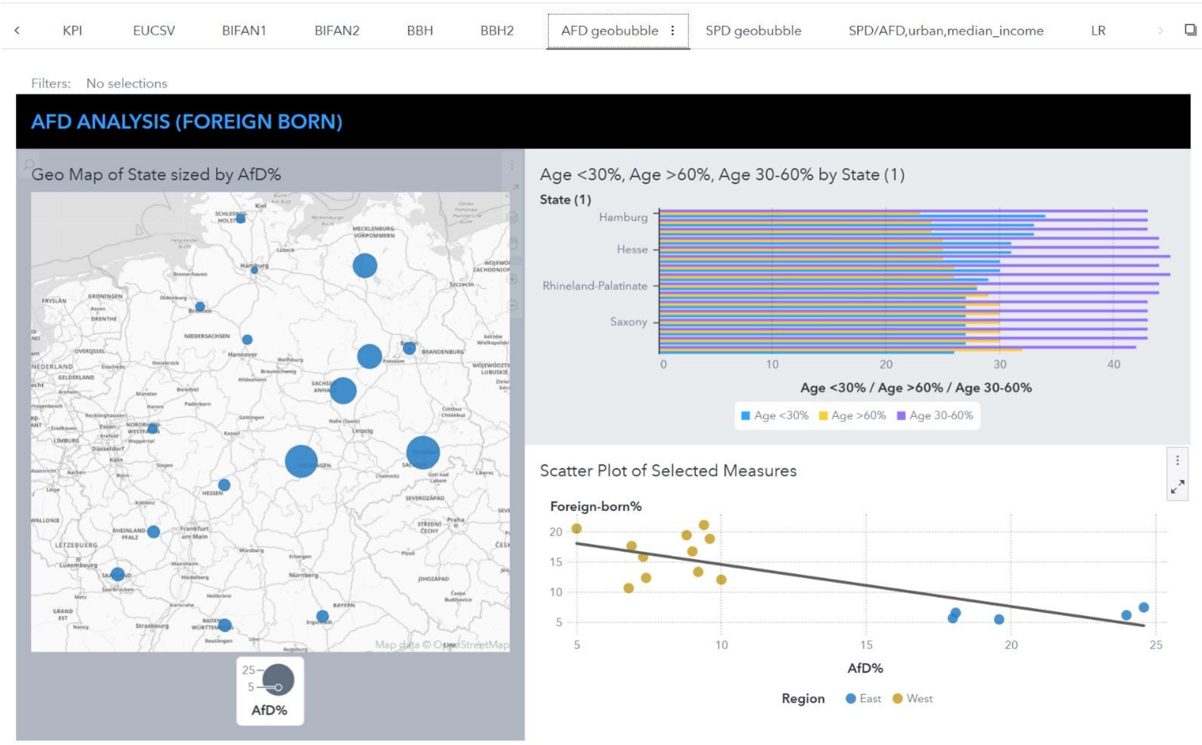



KPI
EUCSV
BIFAN1
BIFAN2
BBH
BBH2
AFD geobubble
SPD geobubble
SPD/AFD,urban,median_income
LR
SPD REGION WISE BUBBLE PLOT FILTERED BY LATICE COLUMN (MEDIAN INCOME).
Vertical guide: Values left of the line are below €24,000 median income; values right are above
Urbanisation% by Median Income (€) sized by SPD%
Urbanisation%
100
80
60
40
Median Income (€)
State (1)
24000
Baden-W...
Bavaria
Berlin
Brandenb...
Bremen
Hamburg
Hesse
Lower Sa...
Mecklen...
North Rhi...
Rhinelan...
Saarland
Saxony
Saxony-A...
Schleswig...
Thuringia
SPD%
37
18
Region
East
West

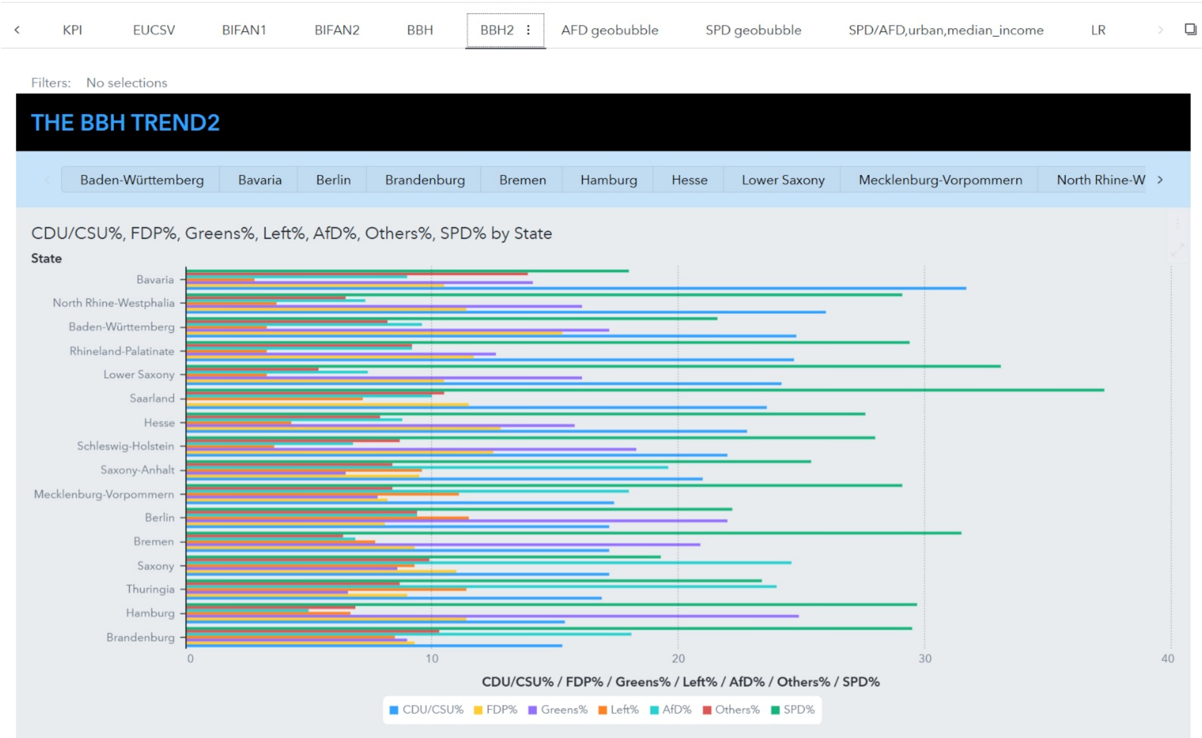

KPI
EUCSV
BIFAN1
BIFAN2
BBH
BBH2
AFD geobubble
SPD geobubble
SPD/AFD,urban,median_income
LR
LINEAR REGRESSION ANALYSIS
Linear Regression of AfD%
Fit: R-Square 0.9705
Observations: 16 of 16
Fit Summary
Region
Unemployment%
Urbanisation%
Median Income (€)
Education% (Tertiary)
0.1
0.01
0.001
0.0001
<0.00001
p-value
Residual Plot
Studentized Deleted Residual
2
1
0
-1
-2
5
10
15
20
25
Predicted Value
Assessment
AfD%
25
20
15
10
5
20
40
60
80
100
Percentile
Predicted Average
Observed Average

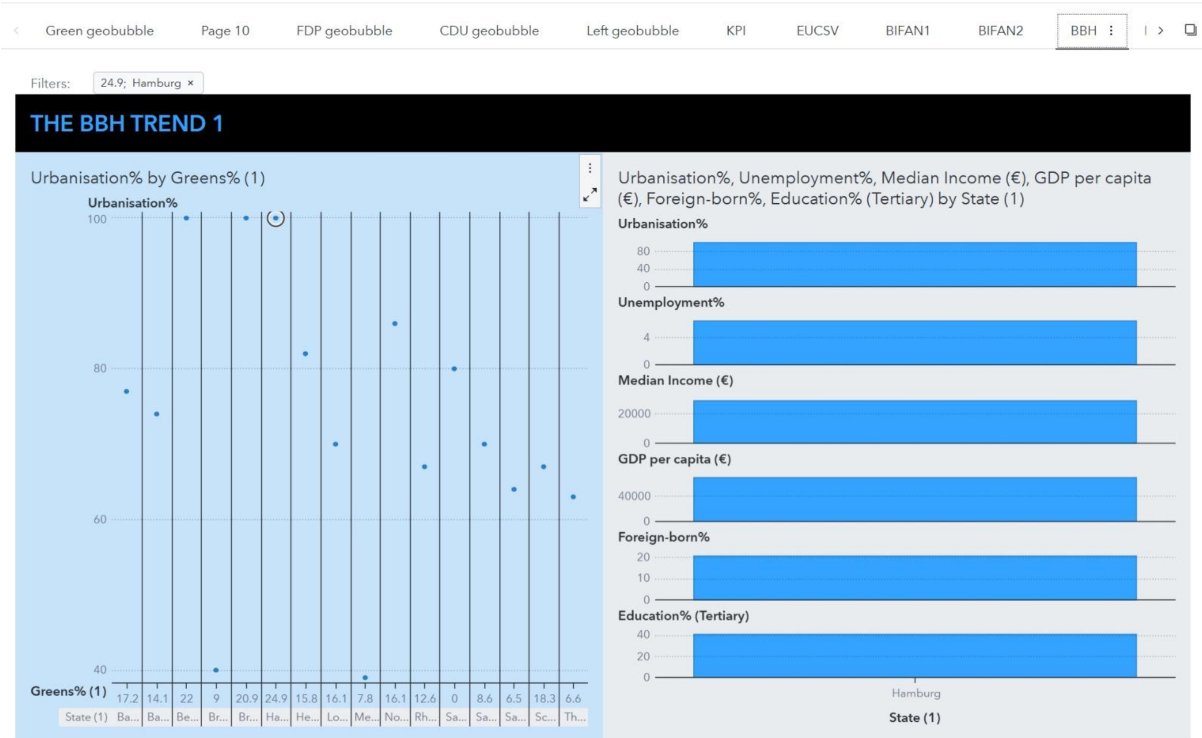


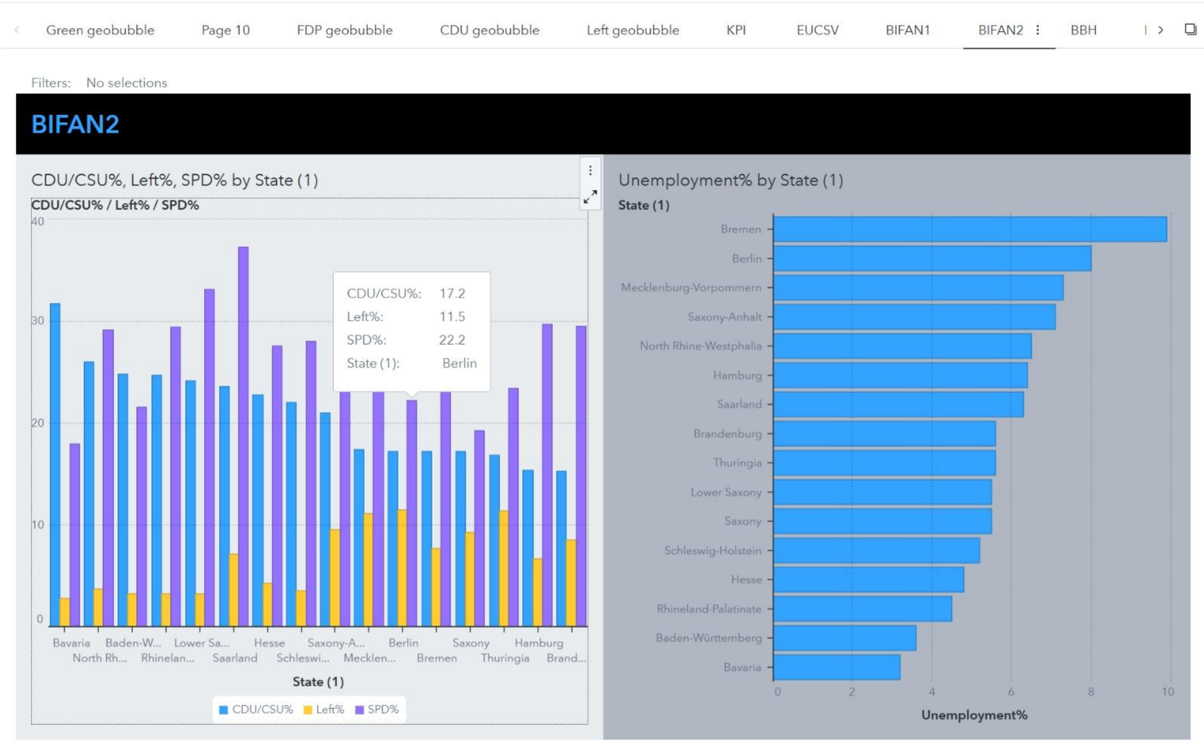
KPI
EUCSV
BIFAN1
BIFAN2
BBH
BBH2
AFD geobubble
SPD geobubble
SPD/AFD,urban,median_income
LR
Filters: No selections
SPD DIRECTLY CORELATION TO UNEMPLOYMENT
Geo Map of State sized by SPD%
37
18
SPD%
Unemployment% by State (1), Frequency
Frequency
2
1
0
Bade...
Bavaria
Berlin
Brande...
Bremen
Hamburg
Hesse
Lower S...
Meckle...
North R...
Rhinela...
Saarland
Saxony
Saxony-...
Schlesw...
Thuri...
State (1)
3.2
9.9
Unemployment%



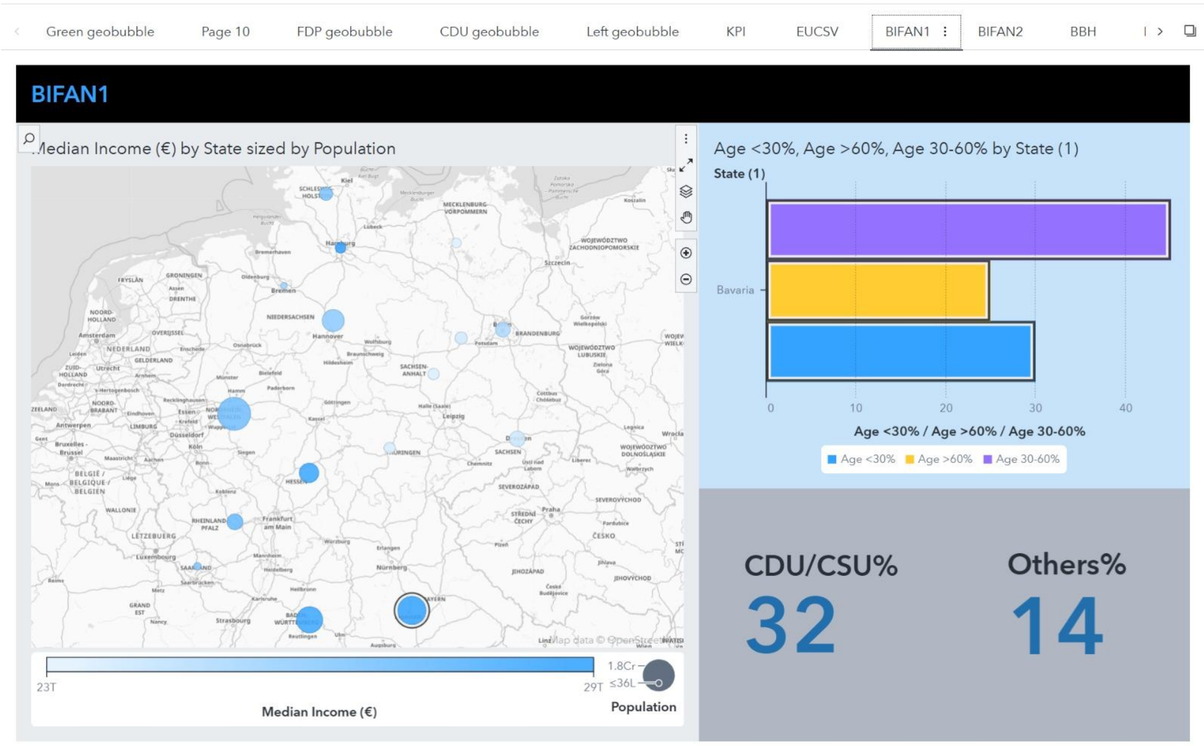
KPI
EUCSV
BIFAN1
BIFAN2
BBH
BBH2
AFD geobubble
SPD geobubble
SPD/AFD,urban,median_income
LR
Filters: No selections
AFD ANALYSIS (FOREIGN BORN)
Geo Map of State sized by AfD%
25
5
AfD%
Age <30%, Age >60%, Age 30-60% by State (1)
State (1)
Hamburg
Hesse
Rhineland-Palatinate
Saxony
0
10
20
30
40
Age <30% / Age >60% / Age 30-60%
Age <30%
Age >60%
Age 30-60%
Scatter Plot of Selected Measures
Foreign-born%
20
15
10
5
5
10
15
20
25
AfD%
Region
East
West


KPI
EUCSV
BIFAN1
BIFAN2
BBH
BBH2
AFD geobubble
SPD geobubble
SPD/AFD,urban,median_income
LR
Filters: No selections
THE BBH TREND2
Baden-Württemberg
Bavaria
Berlin
Brandenburg
Bremen
Hamburg
Hesse
Lower Saxony
Mecklenburg-Vorpommern
North Rhine-W
CDU/CSU%, FDP%, Greens%, Left%, AfD%, Others%, SPD% by State
State
Bavaria
North Rhine-Westphalia
Baden-Württemberg
Rhineland-Palatinate
Lower Saxony
Saarland
Hesse
Schleswig-Holstein
Saxony-Anhalt
Mecklenburg-Vorpommern
Berlin
Bremen
Saxony
Thuringia
Hamburg
Brandenburg
0
10
20
30
40
CDU/CSU% / FDP% / Greens% / Left% / AfD% / Others% / SPD%
CDU/CSU%
FDP%
Greens%
Left%
AfD%
Others%
SPD%


Green geobubble
Page 10
FDP geobubble
CDU geobubble
Left geobubble
KPI
EUCSV
BIFAN1
BIFAN2
BBH
Filters: 24.9; Hamburg ×
THE BBH TREND 1
Urbanisation% by Greens% (1)
Urbanisation%
100
80
60
40
Greens% (1)
17.2
14.1
22
9
20.9
24.9
15.8
16.1
7.8
16.1
12.6
0
8.6
6.5
18.3
6.6
State (1)
Ba...
Ba...
Be...
Br...
Br...
Ha...
He...
Lo...
Me...
No...
Rh...
Sa...
Sa...
Sa...
Sc...
Th...
Urbanisation%, Unemployment%, Median Income (€), GDP per capita (€), Foreign-born%, Education% (Tertiary) by State (1)
Urbanisation%
80
40
0
Unemployment%
4
0
Median Income (€)
20000
0
GDP per capita (€)
40000
0
Foreign-born%
20
10
0
Education% (Tertiary)
40
20
0
Hamburg
State (1)



Green geobubble
Page 10
FDP geobubble
CDU geobubble
Left geobubble
KPI
EUCSV
BIFAN1
BIFAN2
BBH
Filters: No selections
BIFAN2
CDU/CSU%, Left%, SPD% by State (1)
CDU/CSU% / Left% / SPD%
40
30
20
10
0
CDU/CSU%: 17.2
Left%: 11.5
SPD%: 22.2
State (1): Berlin
Bavaria
North Rh...
Baden-W...
Rhinelan...
Lower Sa...
Saarland
Hesse
Schleswi...
Saxony-A...
Mecklen...
Berlin
Bremen
Saxony
Thuringia
Hamburg
Brand...
State (1)
CDU/CSU%
Left%
SPD%
Unemployment% by State (1)
State (1)
Bremen
Berlin
Mecklenburg-Vorpommern
Saxony-Anhalt
North Rhine-Westphalia
Hamburg
Saarland
Brandenburg
Thuringia
Lower Saxony
Saxony
Schleswig-Holstein
Hesse
Rhineland-Palatinate
Baden-Württemberg
Bavaria
0
2
4
6
8
10
Unemployment%

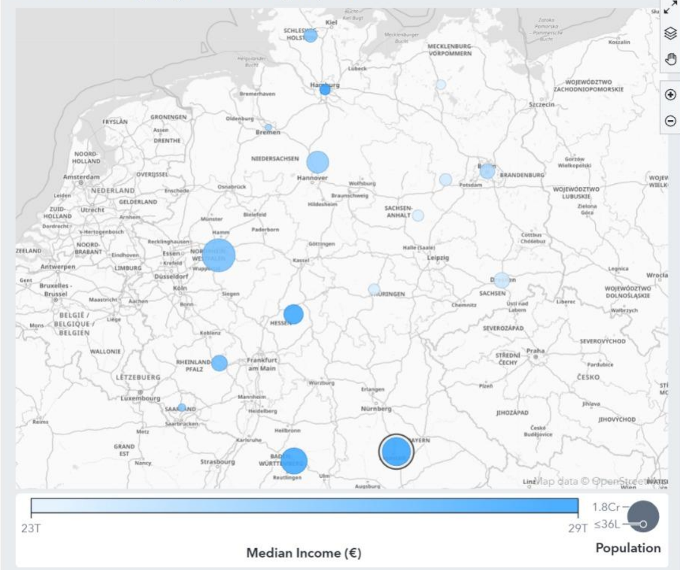


Green geobubble
Page 10
FDP geobubble
CDU geobubble
Left geobubble
KPI
EUCSV
BIFAN1
BIFAN2
BBH
BIFAN1
Median Income (€) by State sized by Population
23T
29T
Median Income (€)
1.8Cr
≤36L
Population
Age <30%, Age >60%, Age 30-60% by State (1)
State (1)
Bavaria
0
10
20
30
40
Age <30% / Age >60% / Age 30-60%
Age <30%
Age >60%
Age 30-60%
CDU/CSU%
32
Others%
14

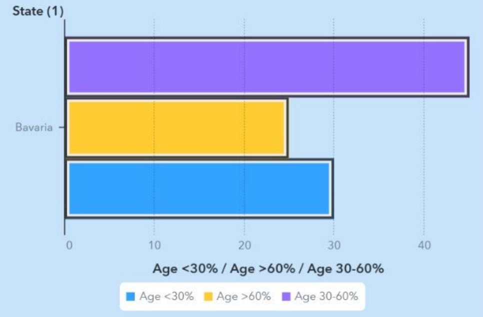
Green geobubble | Page 10 | FDP geobubble | CDU geobubble | Left geobubble | KPI | EUCSV | BIFAN1 | BIFAN2 | BBH

BIFAN1

Median Income (€) by State sized by Population



Age <30%, Age >60%, Age 30-60% by State (1)



CDU/CSU%	Others%
32	14

Green geobubble | Page 10 | FDP geobubble | CDU geobubble | Left geobubble | KPI | EUCSV | BIFAN1 | BIFAN2 | BBH

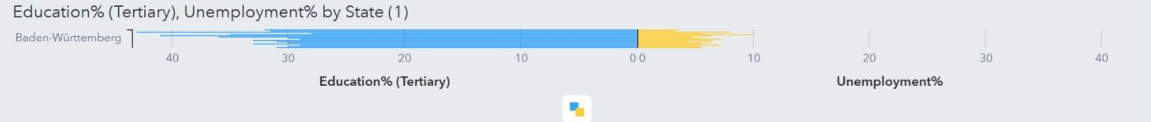
Filters: No selections

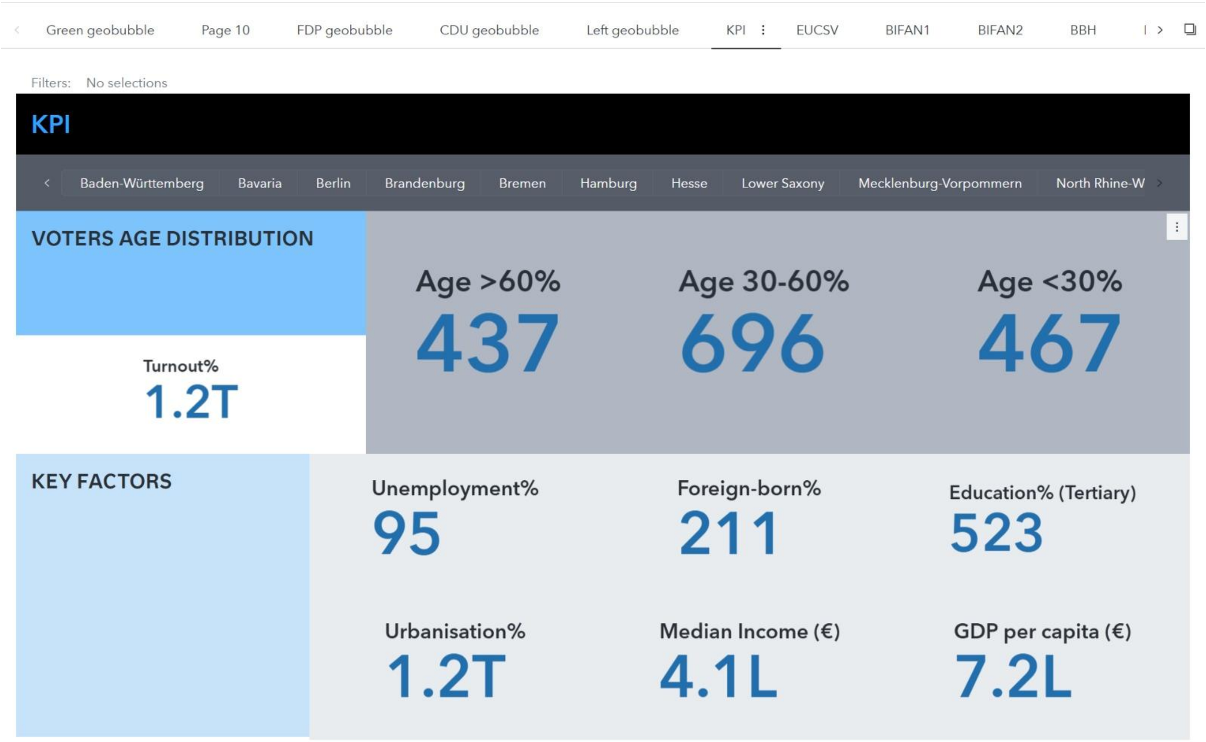
EDUCATION AND UNEMPLOYMENT CORELATION WRT STATES AND VOTE SHARE

Baden-Württemberg	18.9
Bavaria	16.8
Berlin	21.2
Brandenburg	6.6
Bremen	17.7
Hamburg	20.6
Hesse	19.5

AfD% (1) ▲	CDU/CSU% (CAT) ▼	FDP% (1) ▲	Greens% (1) ▲	Left% (1) ▲	Others% (1) ▼	SPD% (1) ▼	State (1) ▼	Frequency
5	15.4	11.4	24.9	6.7	6.9	29.7	Hamburg	1
6.8	22	12.5	18.3	3.6	8.7	28	Schleswig-Holstein	1
6.9	17.2	9.3	20.9	7.7	6.4	31.5	Bremen	1
7.3	26	11.4	16.1	3.7	6.5	29.1	North Rhine-Westphalia	1
7.4	24.2	10.5	16.1	3.3	5.4	33.1	Lower Saxony	1
8.8	22.8	12.8	15.8	4.3	7.9	27.6	Hesse	1
9	31.7	10.5	14.1	2.8	13.9	18	Bavaria	1
9.2	24.7	11.7	12.6	3.3	9.2	29.4	Rhineland-Palatinate	1
9.4	17.2	8.1	22	11.5	9.4	22.2	Berlin	1
9.6	24.8	15.3	17.2	3.3	8.2	21.6	Baden-Württemberg	1

Education% (Tertiary), Unemployment% by State (1)





CONCLUSION/RECOMMENDATION

The analysis demonstrates that socio-economic and demographic differences across Germany's federal states play a significant role in shaping party vote shares. By combining cleaned, enriched, and transformed datasets with regression and statistical techniques, we identified the regions and voter groups most influential in driving electoral outcomes. This approach highlights how targeted socio-demographic factors can explain variations in political support, providing valuable insights for understanding and predicting electoral shifts.

Scope

This study focuses on understanding how socio-economic and demographic factors influence political voting patterns across Germany's 16 federal states. Using indicators such as unemployment, median income, foreign-born population, education level, age distribution, urbanisation and regional divisions, the analysis examines their relationship with vote shares of major parties. The scope is limited to state-level aggregated data from the 2021 federal election and aims to identify the key variables that drive party performance, highlight regional voting behaviours and provide insights that can support evidence-based political strategy and policymaking.

APPENDIX

#References

1. Election and Demographic Data Sources:-

- **Results of the 2021 German federal election(Wikipedia)**
- List of Bundestag constituencies(Wikipedia)
- Bundeswahlleiter - Open Data (Bundestagswahl 2021 open data) for federal election data 2021
- Destatis - Unemployment (regional), GDP(regional), Median income(Regional), Education level(regional)
- Eurostat - regional statistics (optional) for age groups and percentage of foreign-born

2. Tools and Scripts:-

- **Python libraries used for data cleaning, transformation, and analysis (Pandas, Requests).**
- **Web scraping references if you followed a tutorial or documentation (BeautifulSoup).**

#Sources

https://en.wikipedia.org/wiki/List_of_Bundestag_constituencies

https://en.wikipedia.org/wiki/Results_of_the_2021_German_federal_election

Bundeswahlleiter - Open Data (Bundestagswahl 2021 open data) for federal election data 2021

https://www.bundeswahlleiterin.de/bundestagswahlen/2021/ergebnisse/opendata.html

Destatis - Unemployment (regional), GDP(regional), Median income(Regional), Education level(regional)

https://www.destatis.de/EN/Themes/_node.html

Eurostat - regional statistics (optional) for age groups and percentage of foreign-born

https://ec.europa.eu/eurostat